m2sv: A Scalable Benchmark for Map-to-Street-View Spatial Reasoning

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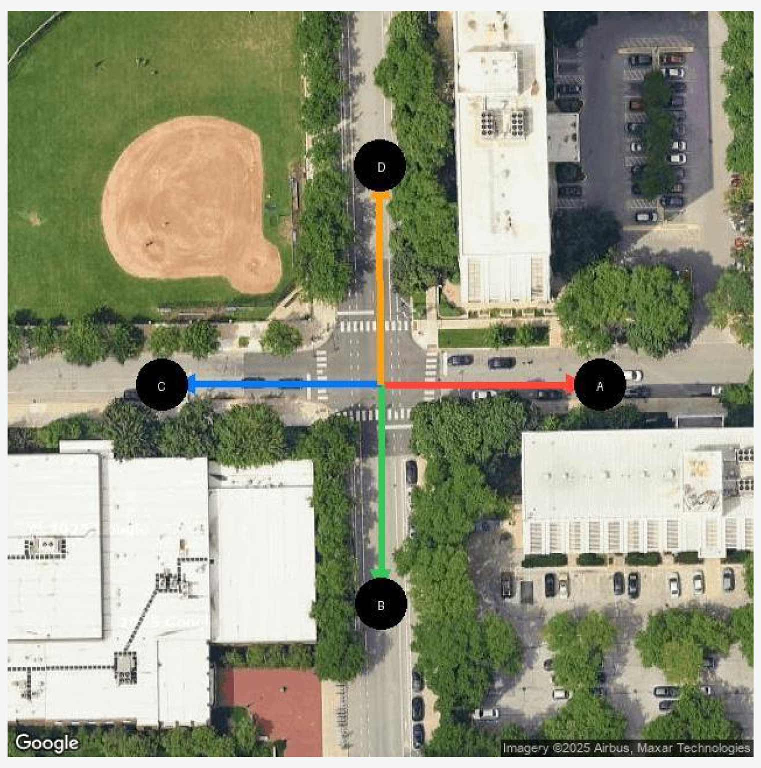
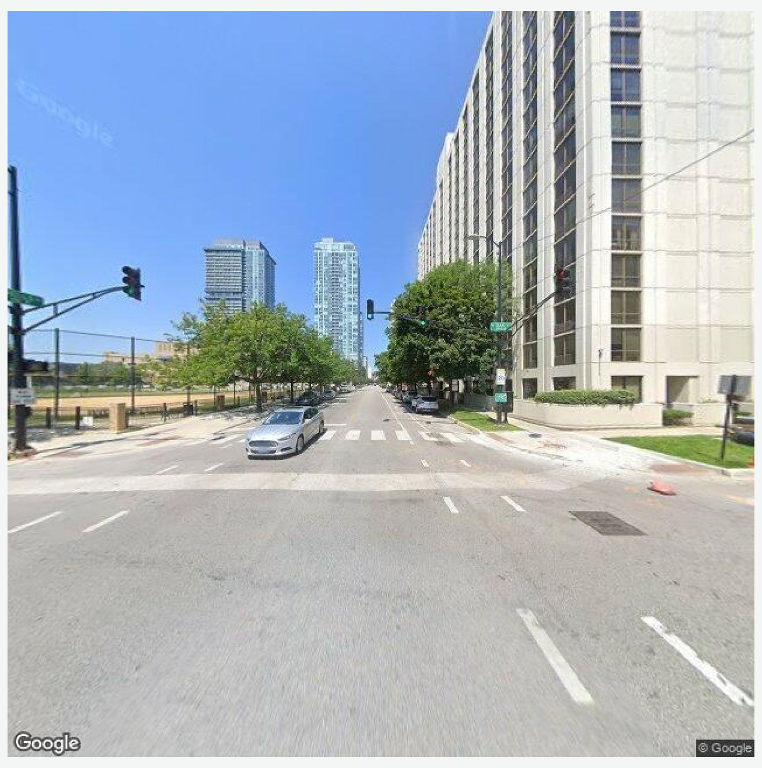
github.com/YosubShin/m2sv   m2sv.yosubshin.com

The Task

Given a **north-up overhead map** with labeled candidate directions and a **Street View photo** taken at the same intersection, infer *which labeled direction the camera was facing*. The camera sits at the intersection center; letters point outward along each road.





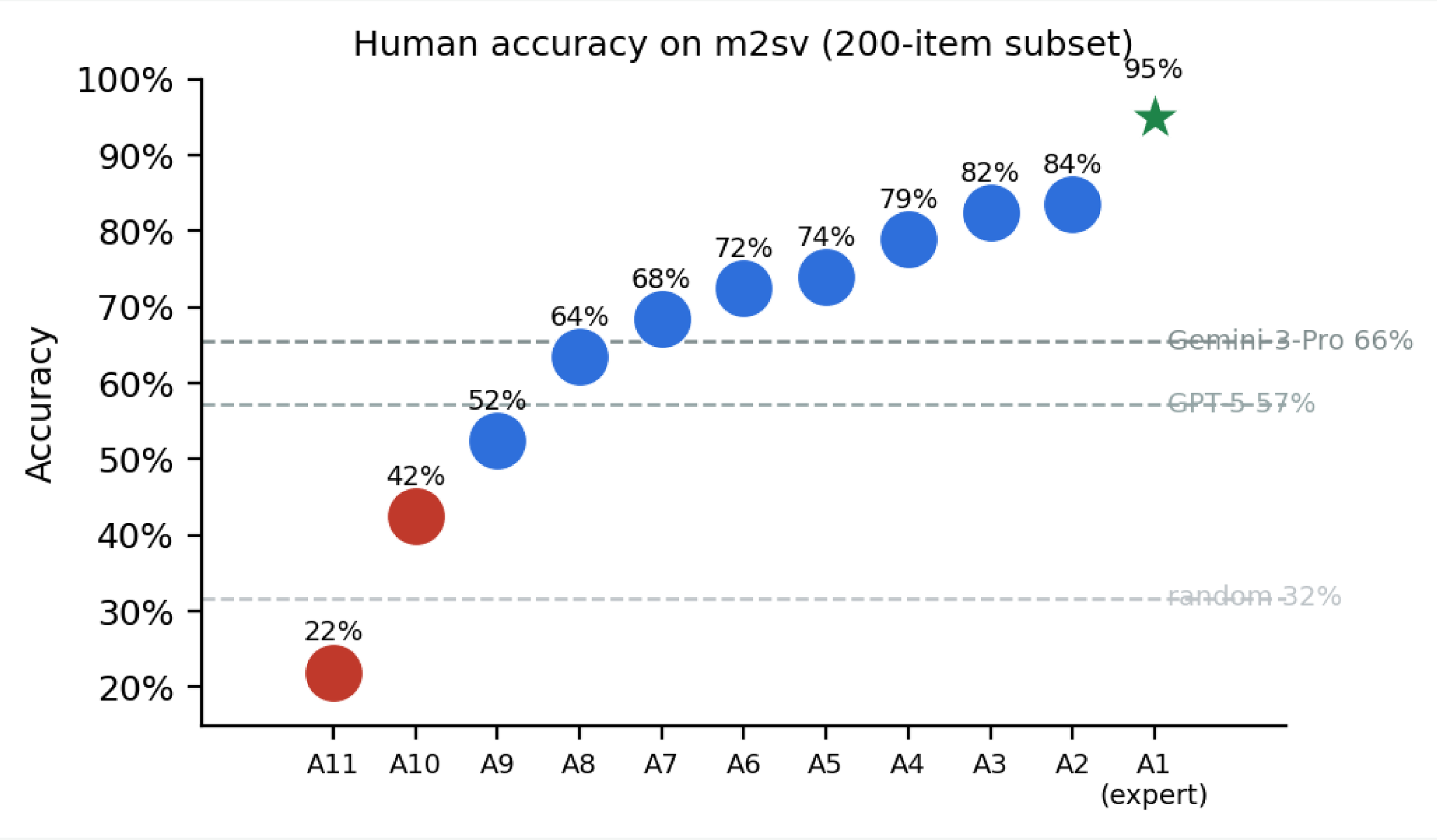
Ground truth: D (camera faces north).

How a correct trace reasons (Gemini-2.5-Pro): *(1)* the photo shows a tall building on the right, an open park/field on the left, and skyscrapers ahead; *(2)* only heading **D** (north) puts a long corner building on the right and a baseball field on the left; *(3)* ⇒ **D**. Solve by aligning *stable* geometry, not transient cues.

Benchmark & Data (open)

- **m2sv-20k** — 20,000 real intersections (10k train / 10k validation) across **32 cities**, multiple continents, controlled candidate ambiguity.
- **m2sv-sft-11k** — 11k curated structured reasoning traces for SFT.
- Released as **blueprints** (coordinates + azimuths) plus rendering code to respect map licensing; all eval/training code public.

Human baseline & agreement



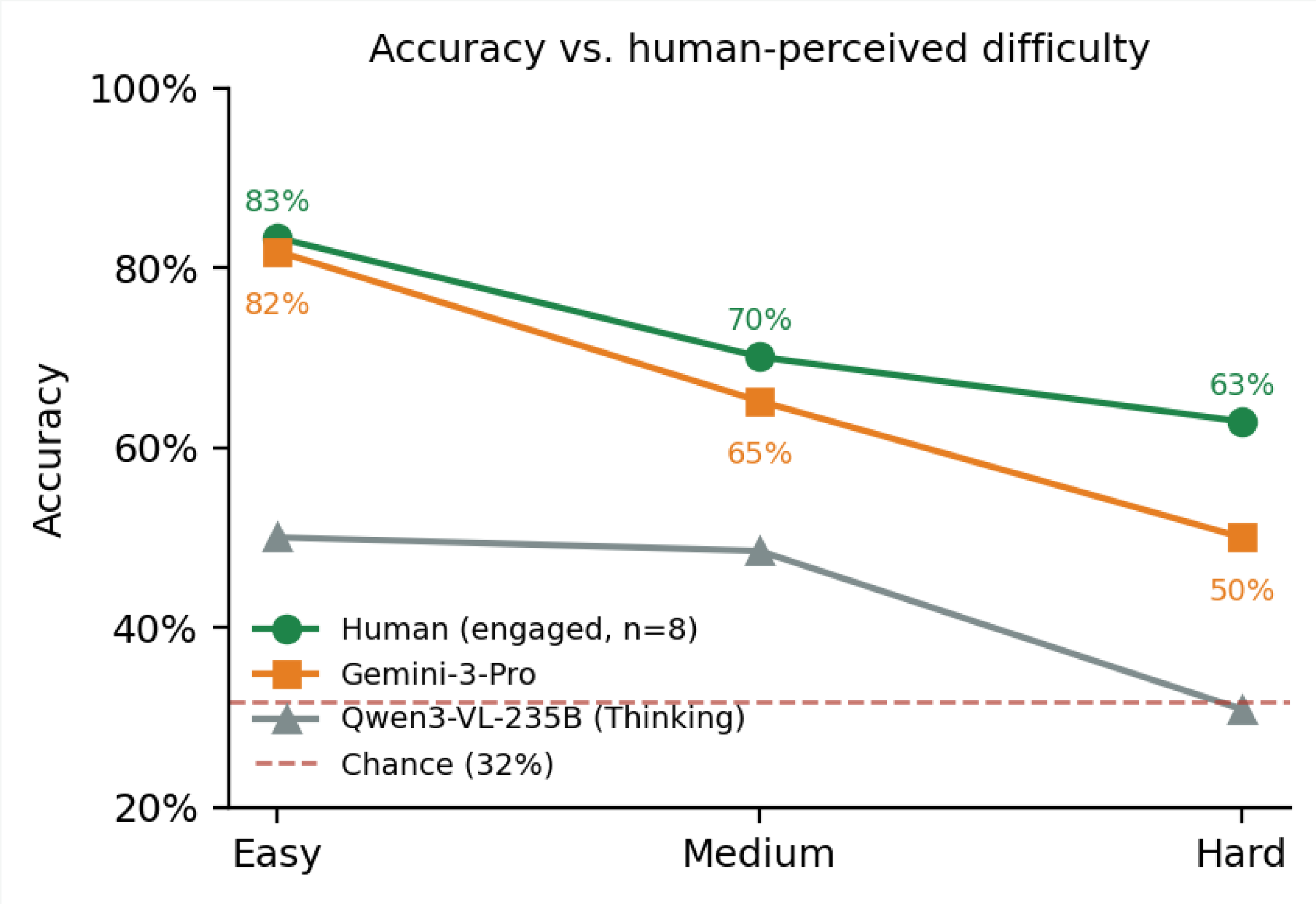
Attentive humans converge (median pairwise $\kappa = 0.48$, up to 0.76); Gemini-3-Pro sits *below* the human consensus (median $\kappa = 0.42$), Qwen3-VL-235B lower still (0.15). The gap shows up in *agreement structure*, not only accuracy.

Zero-shot: humans clear the best VLM

Model	N	Acc. (%)
Gemini-3-Pro	1k	65.2 ± 3.0
GPT-5	1k	57.2 ± 3.1
Gemini-2.5-Pro	1k	47.2 ± 3.1
Qwen3-VL-235B (Think)	1k	42.7 ± 3.1
Random baseline	1k	31.4 ± 2.9
Human (engaged, n=8)	200	**72.0** ± **9.7**
Human (expert)	200	**95.0**

Multi-annotator study (12 participated; the 10 who finished all 200 are analyzed). Two low-agreement outliers excluded by *agreement* ($\kappa_{\text{expert}} < 0.3$), not accuracy.

Difficulty: the hardest-tier collapse



Difficulty bins from **human response time**; accuracy from a model-independent axis. Humans degrade gently (83→63%) and stay above both models; models lose the most ground on the hardest tier, falling to (Qwen) or toward (Gemini, 50%) chance. **Symmetric** intersections (azimuth-gap ratio $S = \frac{\max}{\min}$) are hardest for all models.

Adapting open models (Qwen3-VL-8B, m2sv-20k 10k val)

Model	Training	Acc. (%)
Qwen3-VL-8B	Base	34.3 ± 0.9
Qwen3-VL-8B	SFT	39.8 ± 1.0
Qwen3-VL-8B	SFT+RL	**43.9** ± **1.0**

SFT+RL adds ~10 points on m2sv, but cross-benchmark transfer is weak — fine-tuning induces *benchmark-specific* policies.

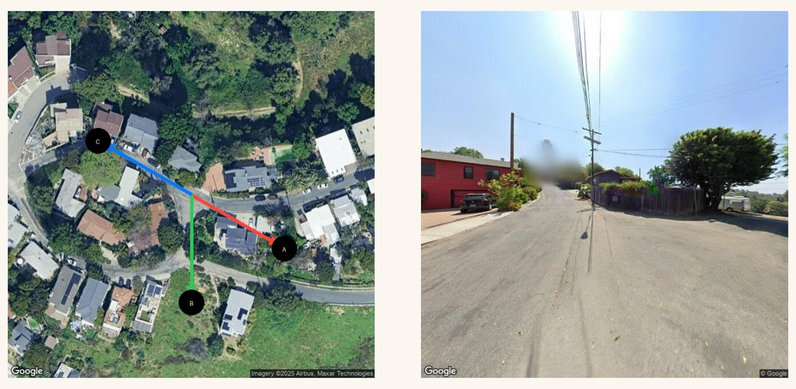
Takeaways

- m2sv isolates a scalable **map↔street-view alignment** primitive frontier VLMs have *not* solved (best 65% vs human 72/95%).
- Humans beat the best model *and* agree with one another; models lie outside that consensus and collapse on the hardest, symmetric cases.
- SFT+RL on m2sv helps (+10 pts) but transfer is limited — failures are about *adapting reasoning to ambiguity*, not missing data.

Seven qualitative failure modes — every case is solved by humans, missed by the model

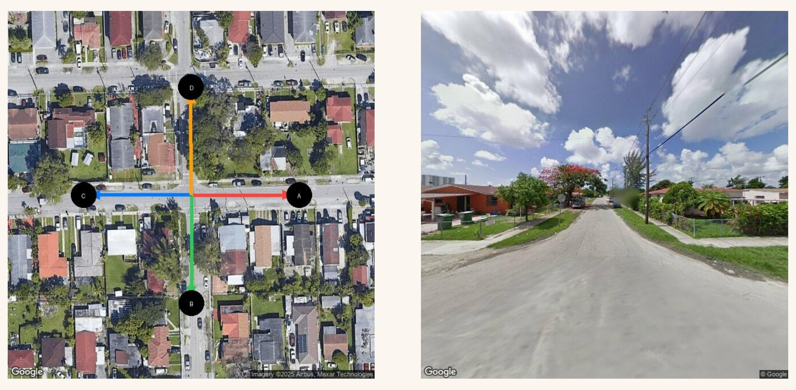
Errors come from *brittle alignment and reliance on unstable evidence*, not missing information. Each card: overhead map, street view, GT/prediction, and the error.

Ego↔allo inversion — GT: A Pred: C



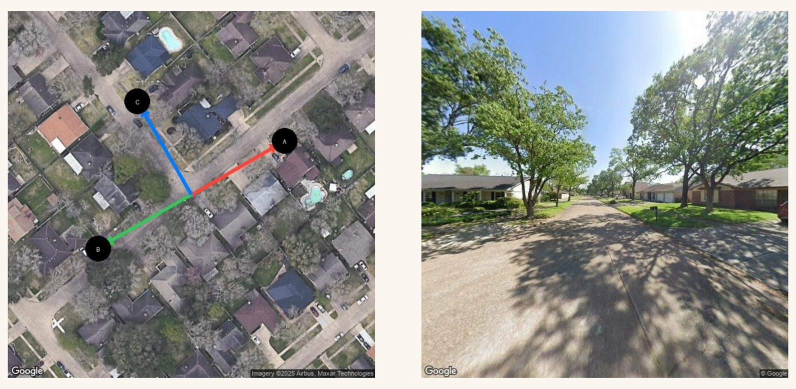
Map shows a *left* curve; the model reasons as if it curved right. Left/right flip at the map↔photo boundary.

Unstable visual cues — GT: B Pred: D



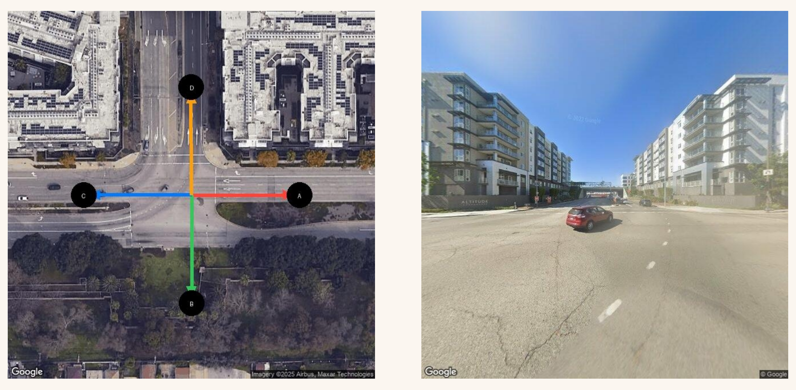
Eliminates on roof color, which is lighting- and angle-dependent and weak in overhead imagery. A human discounts it.

Landmark misbinding — GT: A Pred: B



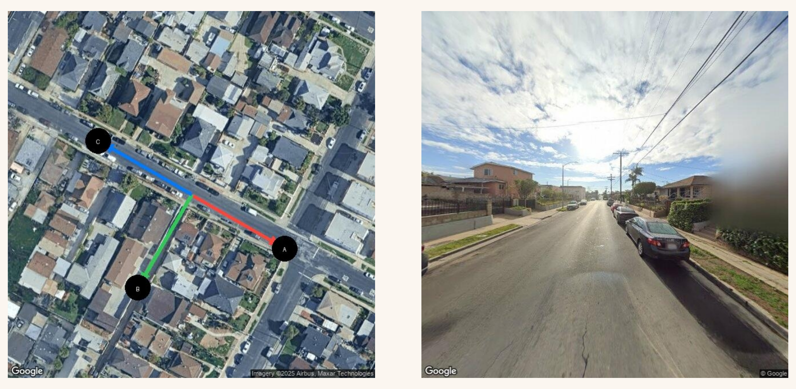
Attributes a pool to the corner house — it is two houses down. Binds a landmark to the wrong spatial anchor.

False landmark — GT: D Pred: A



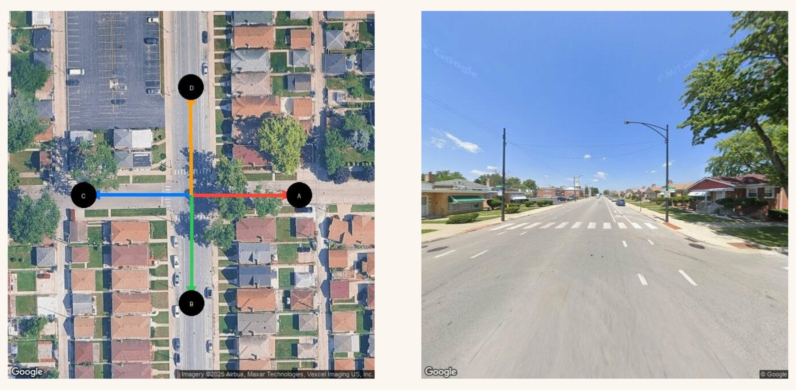
Invents a nonexistent structure (underpass/bridge) and reasons as if it were present in the map.

Spatial inconsistency — GT: A Pred: C



Treats the two sides of one street as having different road widths, instead of enforcing a single coherent layout.

Internal contradiction — GT: D Pred: C



Identifies left–right relations correctly, then contradicts its own assignment mid-reasoning — loses spatial state.

Symmetry trap — GT: D Pred: B

Narrows to two symmetric candidates but will not escalate to fine cues (lane counts, curb geometry) to break the tie.

Supported by NSF NRT-AI 2244574. Contact: yosubs@hawaii.edu · Code/data: github.com/YosubShin/m2sv   Try it: m2sv.yosubshin.com